1 **(a)** Explain why the gravitational potential near to a point mass is negative.

.....

.....

.....

..... [2]

(b) A planet may be assumed to be a uniform sphere. It has gravitational potential ϕ at distance r from the centre of the planet.

The variation with $\frac{1}{r}$ of ϕ is shown in Fig. 1.1.

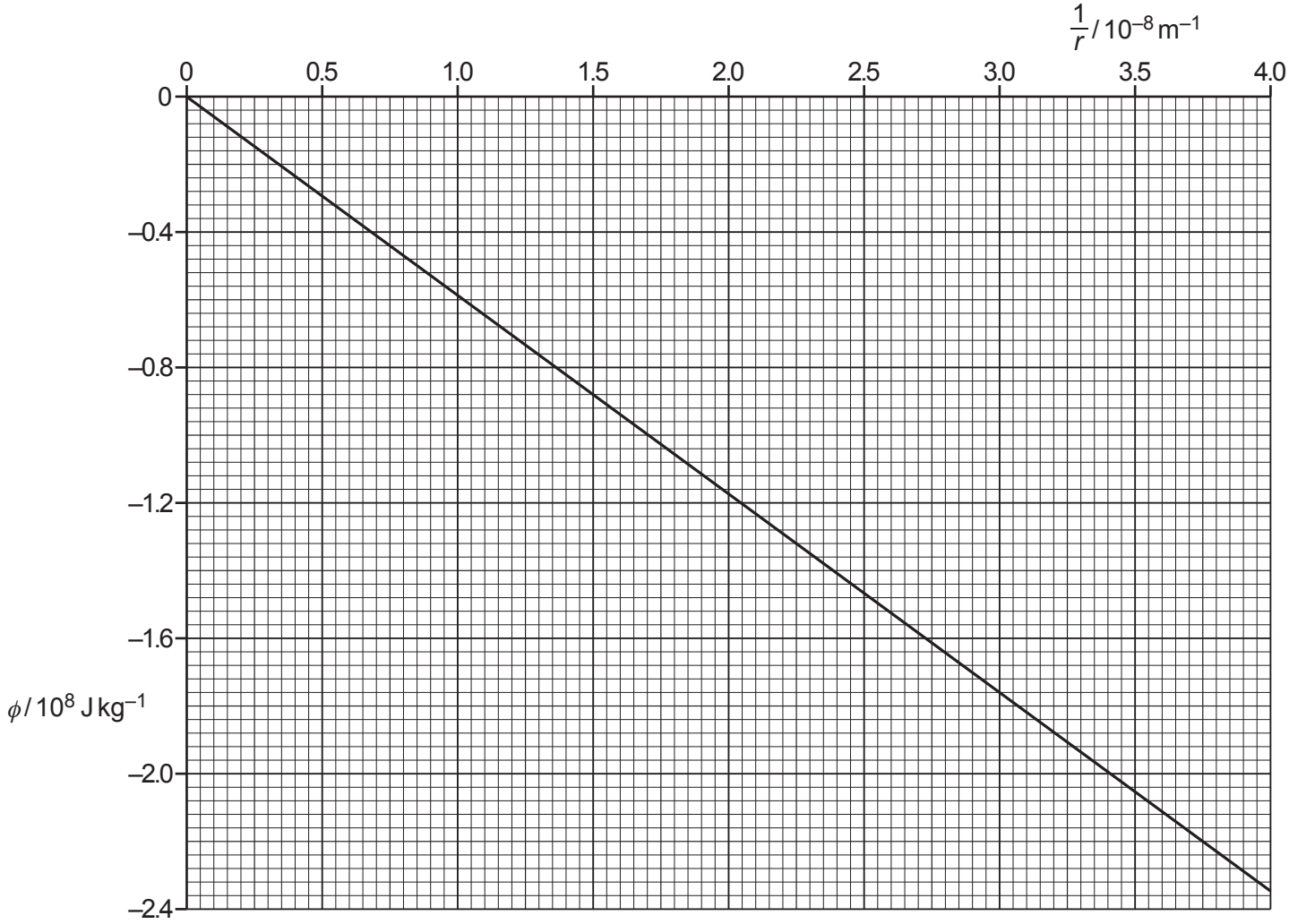


Fig. 1.1

(i) Show that the mass of the planet is 8.8×10^{25} kg.

[2]

(ii) The period of rotation of the planet is 0.72 Earth days.

A satellite in orbit around the planet remains above the same point on the surface of the planet.

Use the mass of the planet in **(b)(i)** to determine the radius R of the orbit of the satellite.

R = ... m [3]

(iii) The speed of the satellite in **(b)(ii)** is 8400 m s^{-1}. The mass of the satellite is 1200 kg.

Determine the additional energy required to move the satellite from its orbit to infinity.

energy required = ... J [3]

[Total: 10]

Markscheme — 9702/42/F/M/24 · Q1

Gravitational potential of a planet — satellite orbit & escape energy

(a) [2 marks]

Gravitational potential is defined to be zero at infinity. The gravitational force is attractive, so work must be done on a mass to move it from a point in the field to infinity — the potential therefore rises towards zero from negative values, so near a point mass it is **negative**.

(b)(i) [2 marks]

$\varphi = -GM/r$, so $M = -\varphi r/G$. From Fig. 1.1, at $1/r = 3.75\times10^{-8}$ m^{-1} the potential is $\varphi = -2.2\times10^{8}$ J kg^{-1}:
$M = (2.2\times10^{8}/3.75\times10^{-8})/6.67\times10^{-11}$ = **8.8×10^{25} kg**

(b)(ii) [3 marks]

$GMm/R^2 = mR\omega^2$ with $\omega = 2\pi/T \rightarrow R^3 = GMT^2/4\pi^2$
$T = 0.72\times24\times3600 = 6.22\times10^{4}$ s
$R^3 = 6.67\times10^{-11}\times8.8\times10^{25}\times(6.22\times10^{4})^2/4\pi^2 = 5.75\times10^{23}$
R = **8.3×10^{7} m**

(b)(iii) [3 marks]

Energy required = ΔGPE − KE = $GMm/R - \frac{1}{2}mv^2$
$= 6.67\times10^{-11}\times8.8\times10^{25}\times1200/(8.3\times10^{7}) - \frac{1}{2}\times1200\times8400^2$
$= 8.49\times10^{10} - 4.23\times10^{10}$ = **4.3×10^{10} J**

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A satellite X, of mass M, orbits a planet at a constant distance $4R$ from the centre of the planet, as shown in Fig. 1.1.

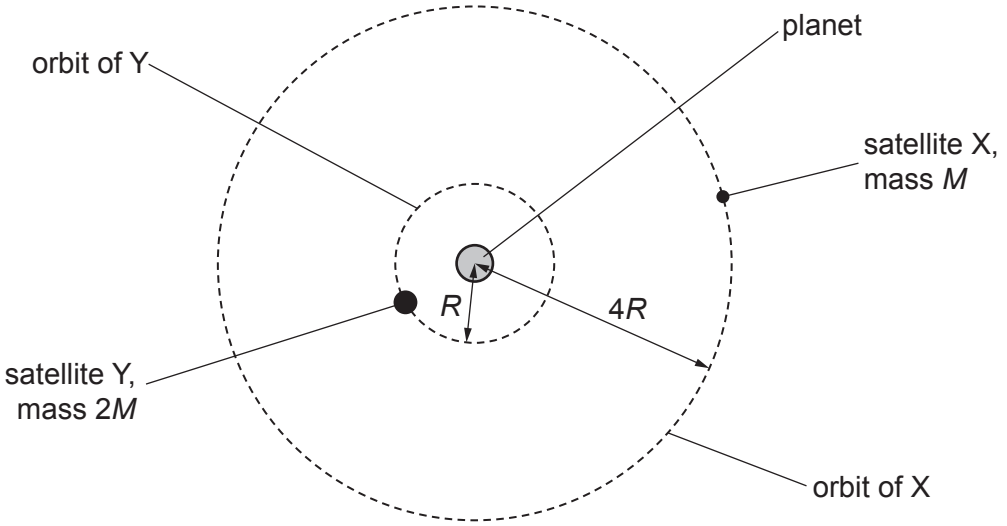


Fig. 1.1 (not to scale)

A second satellite Y, of mass $2M$, orbits the planet with orbital radius R.

The gravitational potential at X due to the planet is $-\Phi$. The planet is a uniform sphere.

(i) Explain why the gravitational potential at X is negative.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ, for the gravitational potential at Y due to the planet.

gravitational potential = ... [2]

(iii) Complete Table 1.1 by giving expressions, in terms of some or all of M, R and Φ, for the quantities indicated for each of the satellites X and Y.

Table 1.1

	satellite X	satellite Y
gravitational field strength at satellite due to planet		
gravitational potential energy of satellite		

[4]

[Total: 10]

Markscheme — 9702/41/M/J/24 · Q1

Satellite in circular orbit — gravitational field strength & energy

(a) [2 marks]

The work done per unit mass
in bringing a small test mass from infinity to the point.

(b)(i) [2 marks]

The gravitational potential is defined as zero at infinity. As the two masses move closer together (or as a mass is brought towards the planet), work is done by the gravitational field, so the potential falls below zero — hence it is negative.

(b)(ii) [2 marks]

$\phi = -GM/r$ and, at X, $-\Phi = -GM/(4R)$, so $GM/R = 4\Phi$.
At Y (radius R): potential = **-4Φ**

(b)(iii) [4 marks]

Since $\Phi = GM/(4R)$, $GM = 4\Phi R$.
Field strength: $g = GM/r^2$
X: $g = GM/(4R)^2$ = **$\Phi/4R$**; Y: $g = GM/R^2$ = **$4\Phi/R$**
Potential energy: $EP = -GMm/r$
X: $EP = -GM \cdot M/4R$ = **$-M\Phi$**; Y: $EP = -GM \cdot 2M/R$ = **$-8M\Phi$**

1 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) A planet may be considered as a uniform sphere.

A satellite is in circular orbit of period T around the planet at a height h above the surface. The height of the orbit can be adjusted by use of the satellite's rocket engines.

Fig. 1.1 shows the variation with h of $T^{\frac{2}{3}}$.

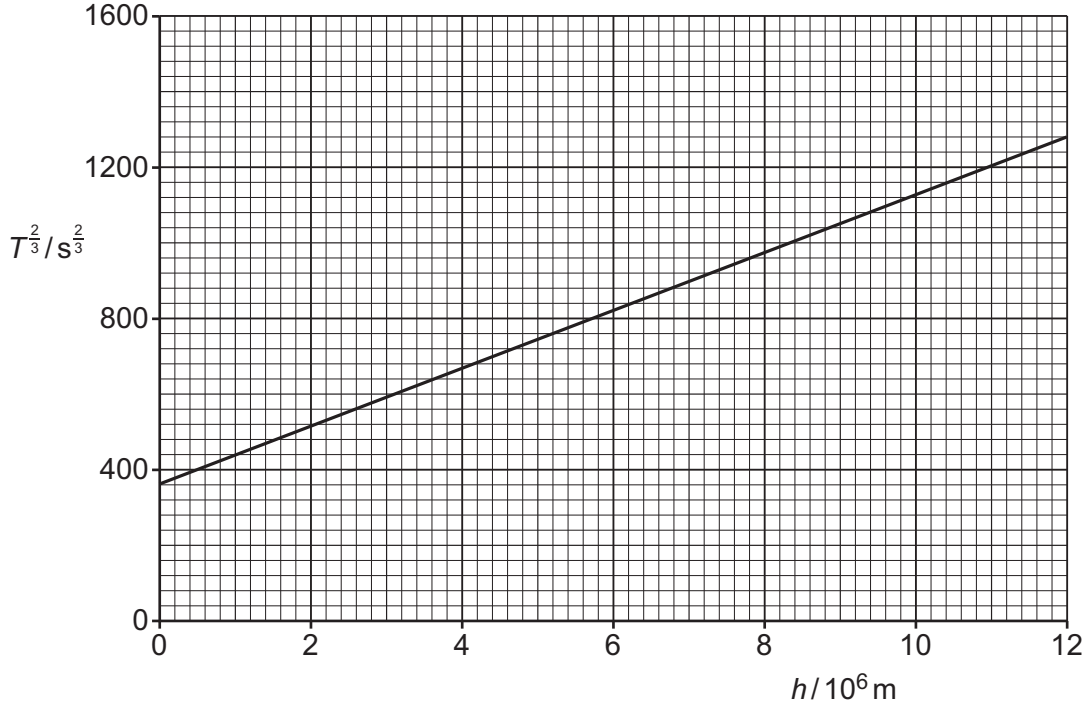


Fig. 1.1

(i) By reference to forces, explain why the orbit of the satellite is circular.

.....

.....

..... [2]

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(ii) Use Newton's law of gravitation to show that h and T are related by

$$(h + B)^3 = \frac{GA}{4\pi^2} T^2$$

where G is the gravitational constant and A and B are constants that depend on the properties of the planet.

[3]

(iii) Use the gradient and intercept of the line in Fig. 1.1 to determine values for A and B. Give units with your answers.

A = ... unit .....

B = ... unit .....

[5]

[Total: 12]

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Markscheme — 9702/41/O/N/24 · Q1

Satellite in circular orbit — Newton's law & orbital period graph

(a) [2 marks]

The gravitational force between two point masses is directly proportional to the product of the masses and inversely proportional to the square of the distance between them: **$F = Gm_1m_2/r^2$**.

(b)(i) [2 marks]

The gravitational force on the satellite acts towards the centre of the planet, so it is always perpendicular to the direction of motion of the satellite. This force provides the **centripetal acceleration**, so the satellite moves in a circular path at constant speed.

(b)(ii) [3 marks]

Gravitational force = centripetal force:
$GMm/(h + B)^2 = m(h + B)\omega^2$, with $\omega = 2\pi/T$
$GM/(h + B)^2 = (h + B)(2\pi/T)^2$
$(h + B)^3 = GMT^2/4\pi^2 =$ **$(GA/4\pi^2)T^2$**
where A = M (the mass of the planet) and B = the radius of the planet.

(b)(iii) [5 marks]

From Fig. 1.1, gradient $m = (1280 - 360)/(12\times10^6) = 7.7\times10^{-5}\ s^{2/3}\ m^{-1}$.
Since $T^{2/3} = m(h + B)$, the intercept is mB, and $m^3 = 4\pi^2/GA$.
$A = 4\pi^2/(Gm^3) = 4\pi^2/(6.67\times10^{-11} \times (7.7\times10^{-5})^3) =$ **1.3×10^{24} kg**
B = intercept/m = $360/(7.7\times10^{-5}) =$ **4.7×10^6 m**

2 The Sun may be considered as a uniform sphere with a mass of 1.99×10^{30} kg and a surface temperature of 5780 K.

A probe with a mass of 2.63 kg moves in a straight line towards the Sun.
When it is at a distance x from the centre of the Sun, the probe measures the gravitational field strength g due to the Sun and the radiant flux intensity F of radiation from the Sun.

(a) Define gravitational field.

.....

..... [1]

(b) For the position of the probe where $x = 1.47 \times 10^{11}$ m:

(i) calculate g

g = ... N kg^{-1} [2]

(ii) determine the gravitational potential energy E_P of the probe.

E_P = ... J [2]

(c) **(i)** Show that, for any particular value of x, the numerical values of g and F are related by

$$g = \frac{4\pi GM}{L} F$$

where M is the mass of the Sun, L is the luminosity of the Sun and G is the gravitational constant.

[3]

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(ii) Fig. 2.1 shows the variation of g with F.

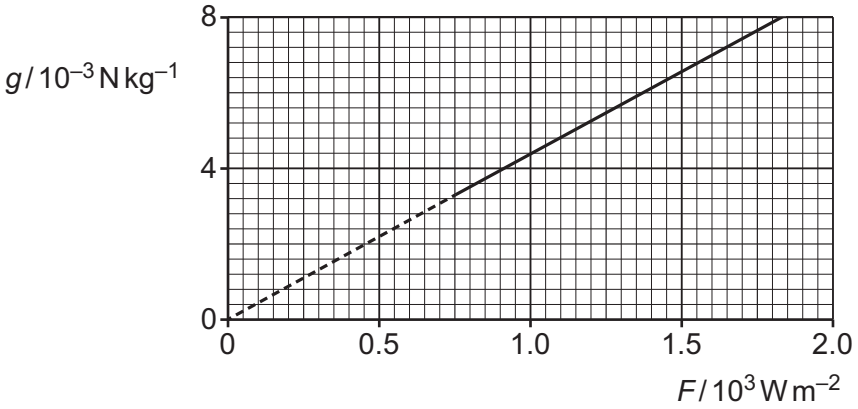


Fig. 2.1

Determine a value for the luminosity L of the Sun. Give a unit with your answer.

L = ... unit ..... [2]

(iii) Use your answer in **(c)(ii)** to determine the radius r of the Sun.

r = ... m [2]

[Total: 12]

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Markscheme — 9702/42/O/N/24 · Q2

Probe approaching the Sun — field strength, potential & luminosity

(a) [1 mark]

Gravitational field (strength) is the **force per unit mass** acting on a small test mass placed in the field.

(b)(i) [2 marks]

$g = GM/x^2 = (6.67\times10^{-11} \times 1.99\times10^{30})/(1.47\times10^{11})^2$
$= \mathbf{6.14\times10^{-3}\ N\ kg^{-1}}$

(b)(ii) [2 marks]

$EP = -GMm/x = -(6.67\times10^{-11} \times 1.99\times10^{30} \times 2.63)/(1.47\times10^{11})$
$= \mathbf{-2.37\times10^{9}\ J}$

(c)(i) [3 marks]

$F = L/(4\pi x^2)$, so $x^2 = L/(4\pi F)$, and $g = GM/x^2$, so $x^2 = GM/g$.
Eliminating x^2: $GM/g = L/(4\pi F)$, giving $\mathbf{g = 4\pi GMF/L}$.

(c)(ii) [2 marks]

Read from Fig. 2.1 a consistent pair of values, e.g. $F = 1.83\times10^{3}$ W m^{-2} when $g = 8.0\times10^{-3}$ N kg^{-1}.
$L = 4\pi GMF/g = (4\pi \times 6.67\times10^{-11} \times 1.99\times10^{30} \times 1.83\times10^{3})/(8.0\times10^{-3})$
$= \mathbf{3.8\times10^{26}\ W}$

(c)(iii) [2 marks]

$L = 4\pi\sigma r^2T^4 \rightarrow r = \sqrt{(L/(4\pi\sigma T^4))}$
$= \sqrt{(3.8\times10^{26}/(4\pi \times 5.67\times10^{-8} \times 5780^4))} = \sqrt{(4.78\times10^{17})}$
$= \mathbf{6.9\times10^{8}\ m}$

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) Mars is a planet that may be considered to be an isolated uniform sphere of radius 3.4×10^6 m.

A satellite of mass 122 kg is in orbit around Mars at a constant height of 1.7×10^6 m above the surface of the planet.

The height of the orbit is increased to 6.8×10^6 m above the surface. This increases the gravitational potential energy of the satellite by 5.1×10^8 J.

(i) Show that the mass of Mars is 6.4×10^{23} kg.

[3]

(ii) Calculate the gravitational potential ϕ at the surface of Mars. Give a unit with your answer.

ϕ = ... unit ..... [2]

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(c) The satellite in **(b)** is moved to an orbit in which the satellite remains at the same point above the surface of Mars.

(i) The orbit has a period of 25 hours.

State what can be deduced from this about the rotation of Mars on its axis.

.....

..... [1]

(ii) State **one** other feature of this orbit.

.....

..... [1]

[Total: 9]

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Markscheme — 9702/41/M/J/25 · Q1

Gravitational potential & field of Mars — satellite orbit

(a) [2 marks]

The **work done per unit mass**
in moving a small test mass **from infinity to the point**.

(b)(i) [3 marks]

The satellite moves from $r_1 = (3.4 + 1.7) \times 10^6 = 5.1 \times 10^6$ m to $r_2 = (3.4 + 6.8) \times 10^6 = 10.2 \times 10^6$ m from the centre of Mars. Using $\Delta E_P = GMm(1/r_1 - 1/r_2)$:
$5.1 \times 10^8 = 6.67 \times 10^{-11} \times M \times 122 \times [(5.1 \times 10^6)^{-1} - (10.2 \times 10^6)^{-1}]$
M = **6.4×10^{23} kg**

(b)(ii) [2 marks]

$\varphi = -GM/R = -(6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(3.4 \times 10^6)$
= **-1.3×10^7 J kg^{-1}**

(c)(i) [1 mark]

The satellite stays over the same point of Mars, so its period equals Mars's rotation period:
Mars takes (just under) 25 hours to rotate once on its axis.

(c)(ii) [1 mark]

The orbit is **equatorial**, and/or the satellite orbits **in the same direction as the rotation of Mars**.

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) Mars is a planet that may be considered to be an isolated uniform sphere of radius 3.4×10^6 m.

A satellite of mass 122 kg is in orbit around Mars at a constant height of 1.7×10^6 m above the surface of the planet.

The height of the orbit is increased to 6.8×10^6 m above the surface. This increases the gravitational potential energy of the satellite by 5.1×10^8 J.

(i) Show that the mass of Mars is 6.4×10^{23} kg.

[3]

(ii) Calculate the gravitational potential ϕ at the surface of Mars. Give a unit with your answer.

ϕ = ... unit ..... [2]

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(c) The satellite in **(b)** is moved to an orbit in which the satellite remains at the same point above the surface of Mars.

(i) The orbit has a period of 25 hours.

State what can be deduced from this about the rotation of Mars on its axis.

.....

..... [1]

(ii) State **one** other feature of this orbit.

.....

..... [1]

[Total: 9]

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Markscheme — 9702/43/M/J/25 · Q1

Gravitational potential & field of Mars — satellite orbit

(a) [2 marks]

The **work done per unit mass**
in moving a small test mass **from infinity to the point**.

(b)(i) [3 marks]

The satellite moves from $r_1 = (3.4 + 1.7) \times 10^6 = 5.1 \times 10^6$ m to $r_2 = (3.4 + 6.8) \times 10^6 = 10.2 \times 10^6$ m from the centre of Mars. Using $\Delta E_P = GMm(1/r_1 - 1/r_2)$:
$5.1 \times 10^8 = 6.67 \times 10^{-11} \times M \times 122 \times [(5.1 \times 10^6)^{-1} - (10.2 \times 10^6)^{-1}]$
M = **6.4×10^{23} kg**

(b)(ii) [2 marks]

$\varphi = -GM/R = -(6.67 \times 10^{-11} \times 6.4 \times 10^{23})/(3.4 \times 10^6)$
= **-1.3×10^7 J kg^{-1}**

(c)(i) [1 mark]

The satellite stays over the same point of Mars, so its period equals Mars's rotation period:
Mars takes (just under) 25 hours to rotate once on its axis.

(c)(ii) [1 mark]

The orbit is **equatorial**, and/or the satellite orbits **in the same direction as the rotation of Mars**.

1 **(a)** Define gravitational field.

.....

..... [1]

(b) The gravitational field strength g at a distance x from the centre of a uniform spherical planet of mass M is given by the expression

$$g = \frac{GM}{x^2}$$

where G is the gravitational constant and distance x is greater than the radius of the planet.

(i) Describe the pattern of the field lines outside the planet that represent the gravitational field due to the planet.

.....

.....

..... [2]

(ii) Explain why, for small changes in vertical height near the surface of the planet, g may be assumed to be constant.

.....

.....

..... [2]

(c) Assume that the Earth is a uniform sphere. For the Earth, the product GM is equal to $3.99 \times 10^{14}\,m^3\,s^{-2}$.

(i) Determine a value, to three significant figures, for the radius R of the Earth.

R = ... m [2]

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(ii) Calculate the gravitational potential at the Earth's surface. Give a unit with your answer.

gravitational potential = ... unit ..... [2]

(d) Explain why the gravitational potential energy of two point masses is always negative.

.....

.....

..... [2]

[Total: 11]

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Markscheme — 9702/44/M/J/25 · Q1

Gravitational field & potential of the Earth

(a) [1 mark]

Gravitational field is the **force per unit mass** acting on a small test mass placed at the point.

(b)(i) [2 marks]

The field lines are **radial**, pointing **towards the centre of the planet**.

(b)(ii) [2 marks]

For small changes in height, the change in height is **very much smaller than the radius** of the planet, so $(R + h)^2 \approx R^2$ — the field lines are approximately parallel and g is approximately constant.

(c)(i) [2 marks]

At the surface $g = GM/R^2$, so $9.81 = 3.99 \times 10^{14}/R^2$
$R = \sqrt{(3.99 \times 10^{14}/9.81)} = \mathbf{6.38 \times 10^6\ m}$

(c)(ii) [2 marks]

$\varphi = -GM/R = -3.99 \times 10^{14}/(6.38 \times 10^6)$
$= \mathbf{-6.25 \times 10^7\ J\ kg^{-1}}$

(d) [2 marks]

The potential energy is **zero at infinite separation**, and the gravitational force is **attractive** — work is done by the field as the masses come together, so the potential energy falls below zero and is always negative.

1 **(a)** State Newton's law of gravitation.

.....

.....

..... [2]

(b) A binary star consists of star A, of mass 4.0×10^{30} kg, and star B, of mass 2.0×10^{30} kg, separated by a distance of 3.3×10^{12} m. The stars are both in circular orbit around their common centre of gravity X, as shown in Fig. 1.1.

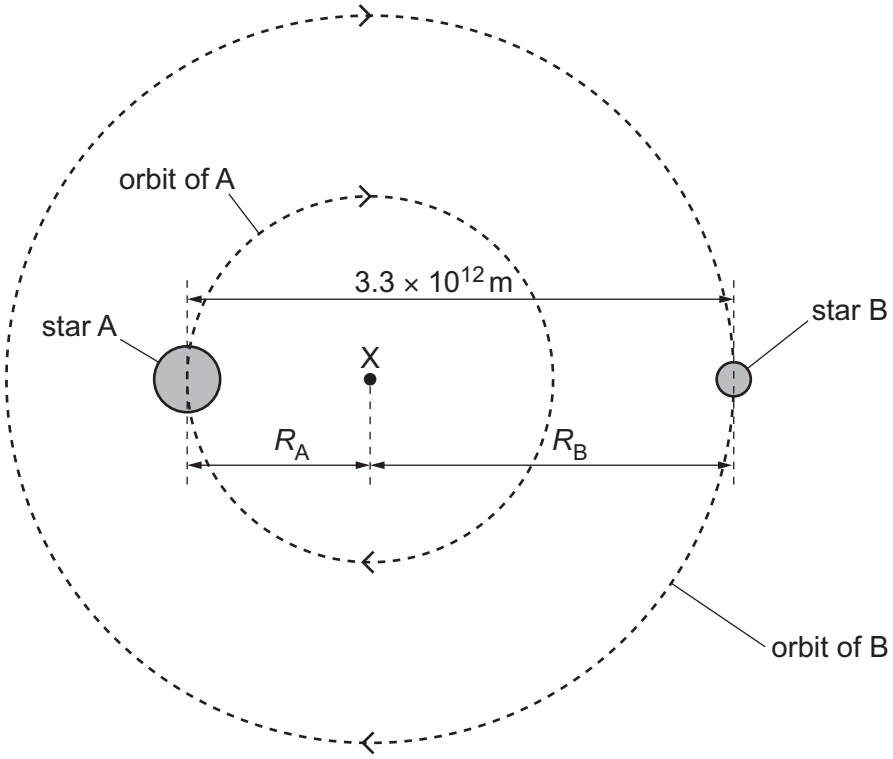


Fig. 1.1

The radius R_B of the orbit of star B is double the radius R_A of the orbit of star A.

(i) Use Newton's law of gravitation to calculate the magnitude of the gravitational force exerted by each star on the other.

force = ... N [2]

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(ii) Calculate the centripetal acceleration of star A.

acceleration = ... $m\,s^{-2}$ [1]

(iii) Use your answer in **(b)(ii)** to determine the period of the orbit of star A.

period = ... s [3]

(iv) By placing a tick (✓) in each row, complete Table 1.1 to show how the quantities indicated for star B compare with the same quantities for star A.

Table 1.1

	B less than A	B equal to A	B greater than A
centripetal acceleration			
linear speed			
period			

[3]

[Total: 11]

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Markscheme — 9702/44/O/N/25 · Q1

Binary star — Newton’s law, orbit & centripetal acceleration

(a) [2 marks]

The gravitational force is **(directly) proportional to the product of the two masses** and **inversely proportional to the square of their separation**.

(b)(i) [2 marks]

$F = Gm_1m_2/r^2 = 6.67 \times 10^{-11} \times 4.0 \times 10^{30} \times 2.0 \times 10^{30}/(3.3 \times 10^{12})^2$
$= \mathbf{4.9 \times 10^{25}\ N}$

(b)(ii) [1 mark]

$a = F/m = 4.9 \times 10^{25}/4.0 \times 10^{30}$
$= \mathbf{1.2 \times 10^{-5}\ m\ s^{-2}}$

(b)(iii) [3 marks]

$a = r\omega^2$ with $\omega = 2\pi/T$. Since $R_B = 2R_A$ and $R_A + R_B = 3.3 \times 10^{12}$ m, $R_A = 3.3 \times 10^{12}/3 = 1.1 \times 10^{12}$ m.
$T = 2\pi\sqrt{(r/a)} = 2\pi\sqrt{(1.1 \times 10^{12}/1.2 \times 10^{-5})}$
$= \mathbf{1.9 \times 10^{9}\ s}$

(b)(iv) [3 marks]

centripetal acceleration: **B greater than A** ($a = F/m$, and B has the smaller mass); linear speed: **B greater than A** ($v = r\omega$, and B has the larger radius); period: **B equal to A** (both orbit X with the same period).

2 **(a)** The magnitude of the gravitational potential on the surface of a planet of radius R is ϕ. The planet can be considered to be an isolated sphere.

On Fig. 2.1, sketch the variation of the gravitational potential with distance x from the centre of the planet for values of x between R and $4R$.

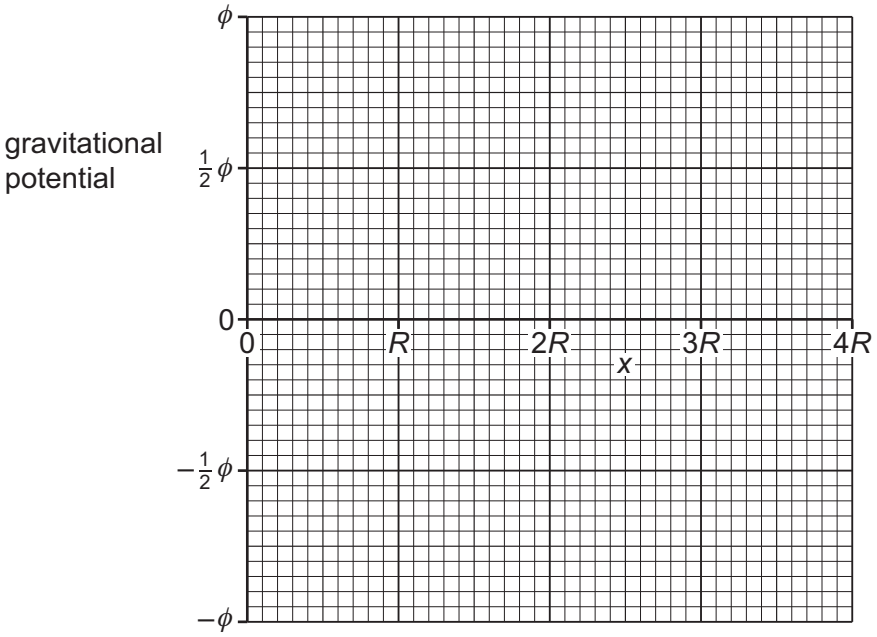


Fig. 2.1

[3]

(b) A satellite is in a geostationary orbit above the Earth. At time t = 0, the magnitude of the gravitational potential due to the Earth at the location of the satellite is ϕ.

On Fig. 2.2, sketch the variation of the gravitational potential due to the Earth at the location of the satellite for values of t between t = 0 and t = 24 hours.

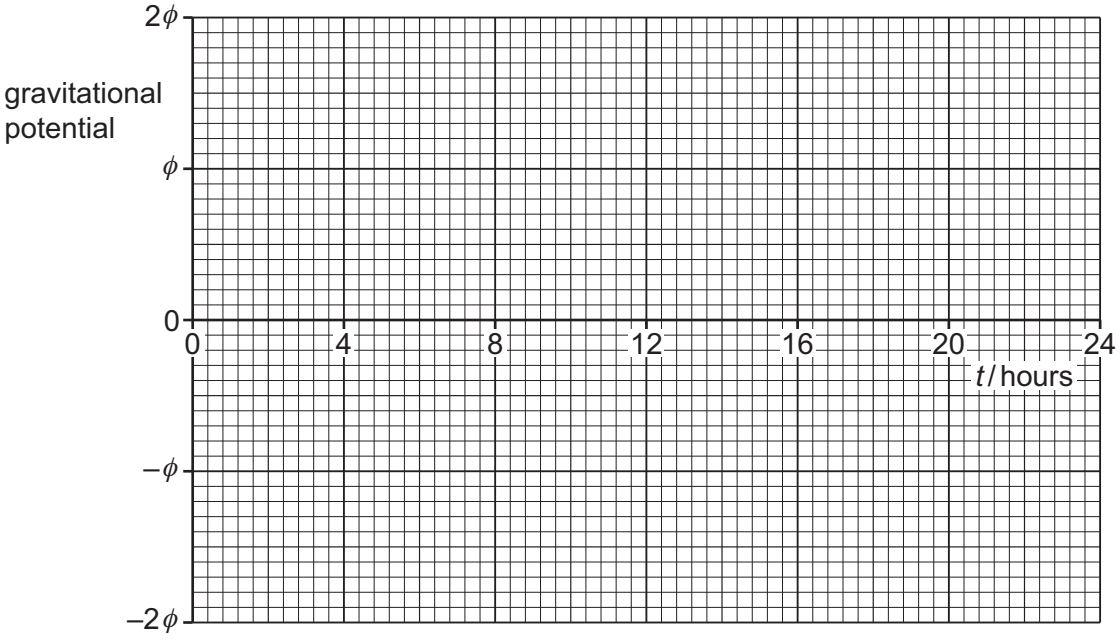


Fig. 2.2

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(c) The electric potential difference (p.d.) between two parallel plates is V, as shown in Fig. 2.3.

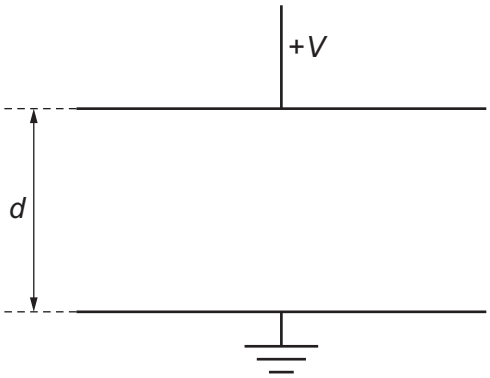


Fig. 2.3

The distance between the plates is d. The region between the plates is a vacuum.

On Fig. 2.4, sketch the variation of the electric potential with distance from the positive plate.

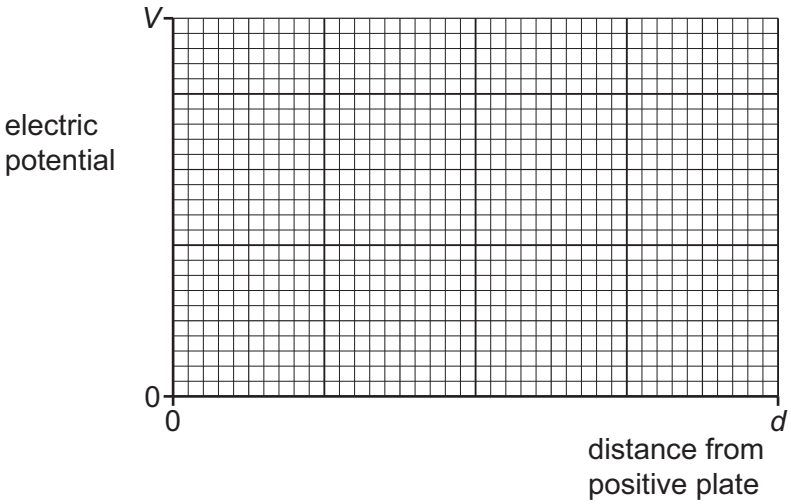


Fig. 2.4

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Markscheme — 9702/42/F/M/25 · Q2

Gravitational potential of a planet & electric potential between plates

(a) [3 marks]

Sketch φ against x: a curve lying **entirely in the negative-φ region** for x between R and 4R, starting at (R, −φ) with a **continuously decreasing magnitude and continuously decreasing gradient** (concave up towards zero), and passing through **(2R, −½φ)** and **(4R, −¼φ)**. The potential is φ = −GM/x, so it halves as x doubles.

(b) [2 marks]

A geostationary satellite stays at the **same point** above the Earth, so its gravitational potential is constant: a **horizontal straight line from t = 0 to t = 24 hours**, starting at (0, −φ).

(c) [2 marks]

The field between the plates is uniform, so the potential falls uniformly: a **straight line with constant negative gradient from (0, V) to (d, 0)**, i.e. the potential decreases linearly from the positive plate to the negative plate.

3 **(a)** Define gravitational field at a point.

.....

..... [1]

(b) Fig. 3.1 shows an isolated point mass of mass *M*.

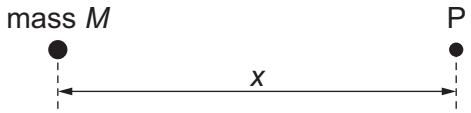


Fig. 3.1

Point P is at distance *x* from the point mass.

(i) By considering the force exerted by the point mass on a test mass of mass *m* placed at P, derive an equation for the gravitational field strength *g* at P, in terms of *M* and *x*. Identify any other symbols you use.

[2]

(ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

(iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]

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(c) Two identical isolated uniform spheres X and Y each have radius R. The centres of the spheres are separated by distance L, as shown in Fig. 3.3.

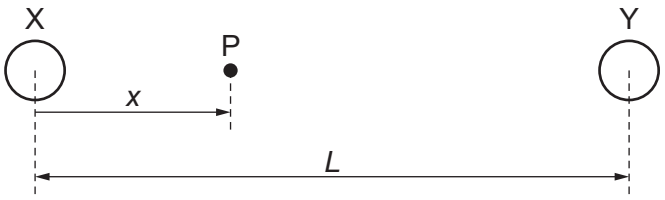


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.
The gravitational field strength at the surface of each sphere is g_0.

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

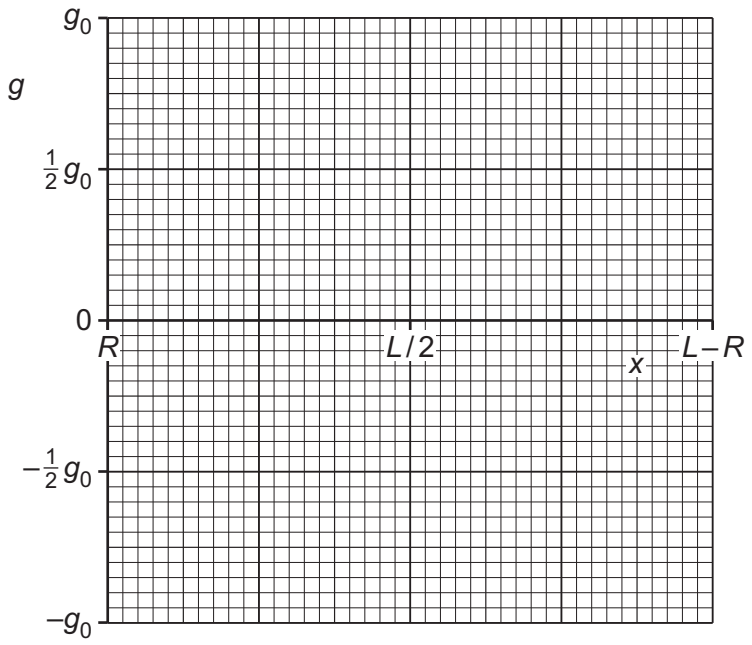


Fig. 3.4

[3]

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Markscheme — 9702/41/O/N/25 · Q3

Gravitational field — point masses & two spheres

(a) [1 mark]

The **force per unit mass** acting on a small test mass placed at the point.

(b)(i) [2 marks]

$F = GMm/x^2$ and $g = F/m$, so $g = [GMm/x^2]/m = \mathbf{GM/x^2}$, where G is the gravitational constant.

(b)(ii) [1 mark]

An arrow at P pointing **directly towards the point mass**.

(b)(iii) [2 marks]

The field at Q is in the **opposite direction** to that at P (towards the mass), and is **four times the field strength at P** (Q is at distance x/2, and $g \propto 1/x^2$).

(c) [3 marks]

A line starting at $(R, -g_0)$ and ending at $(L - R, +g_0)$, passing through $(L/2, 0)$, becoming shallower from R to L/2 and steeper again from L/2 to L − R.

3 **(a)** Define gravitational field at a point.

.....

..... [1]

(b) Fig. 3.1 shows an isolated point mass of mass *M*.

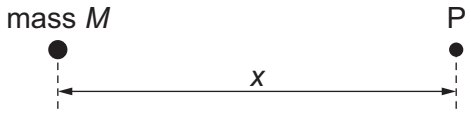


Fig. 3.1

Point P is at distance *x* from the point mass.

(i) By considering the force exerted by the point mass on a test mass of mass *m* placed at P, derive an equation for the gravitational field strength *g* at P, in terms of *M* and *x*. Identify any other symbols you use.

[2]

(ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

(iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

.....

.....

..... [2]

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(c) Two identical isolated uniform spheres X and Y each have radius R. The centres of the spheres are separated by distance L, as shown in Fig. 3.3.

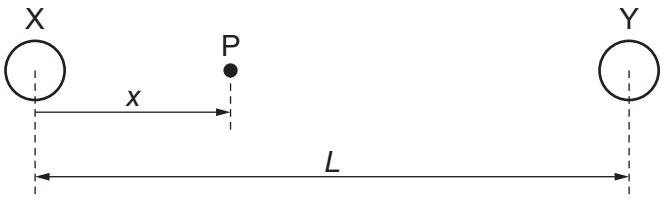


Fig. 3.3

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.
The gravitational field strength at the surface of each sphere is g_0.

On Fig. 3.4, sketch the variation with x of the gravitational field g at point P between $x = R$ and $x = L - R$.

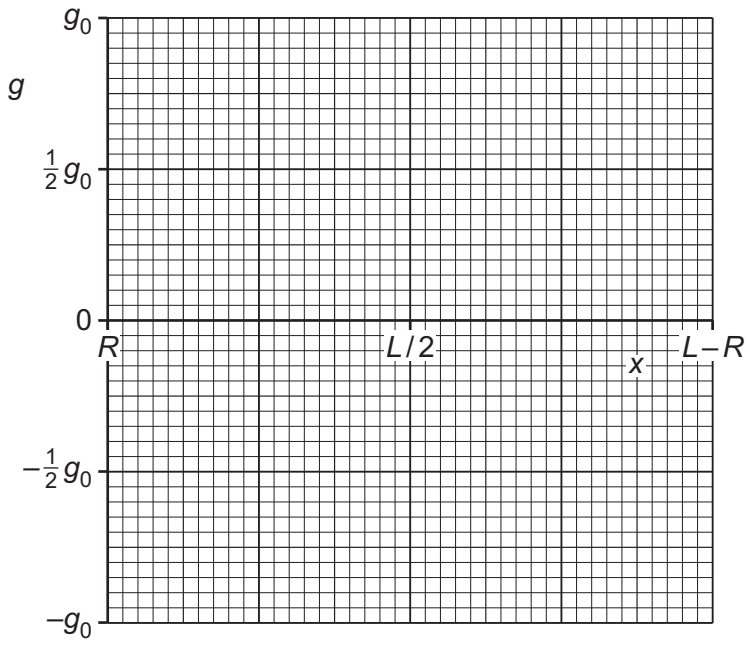


Fig. 3.4

[3]

[Total: 9]

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Markscheme — 9702/43/O/N/25 · Q3

Gravitational field — point masses & two spheres

(a) [1 mark]

The **force per unit mass** acting on a small test mass placed at the point.

(b)(i) [2 marks]

$F = GMm/x^2$ and $g = F/m$, so $g = [GMm/x^2]/m = \mathbf{GM/x^2}$, where G is the gravitational constant.

(b)(ii) [1 mark]

An arrow at P pointing **directly towards the point mass**.

(b)(iii) [2 marks]

The field at Q is in the **opposite direction** to that at P (towards the mass), and is **four times the field strength at P** (Q is at distance x/2, and $g \propto 1/x^2$).

(c) [3 marks]

A line starting at $(R, -g_0)$ and ending at $(L - R, +g_0)$, passing through $(L/2, 0)$, becoming shallower from R to L/2 and steeper again from L/2 to L − R.

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

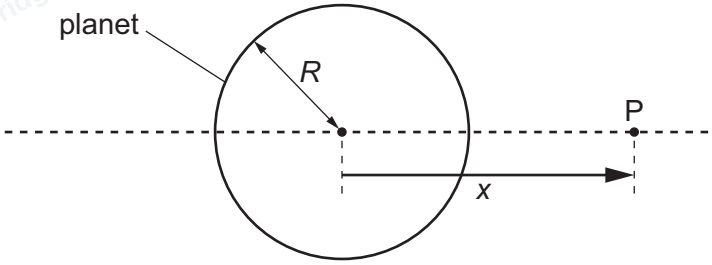


Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

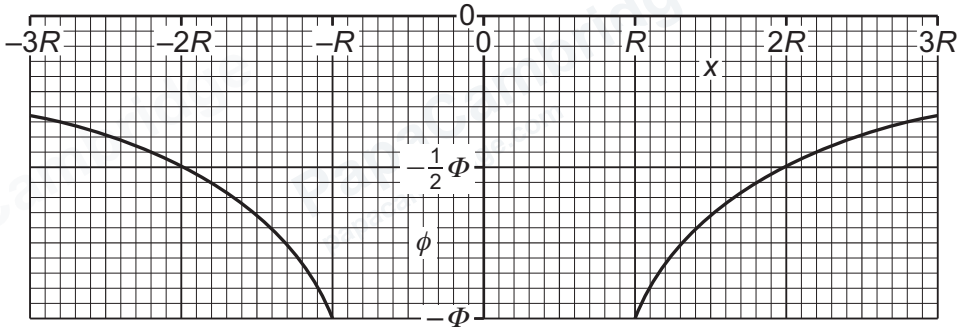


Figure 1.2

(i) Explain why ϕ is negative for all values of x.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use.

M = ... [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet.

g_0 = ... [1]

(iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

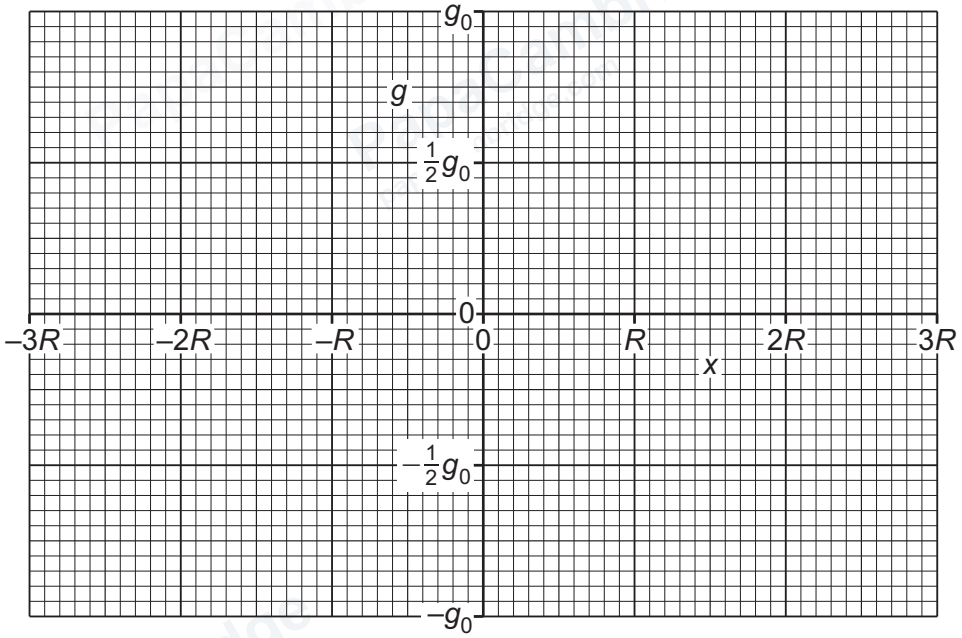


Figure 1.3

[3]

[Total: 10]

Markscheme — 9702/41/M/J/26 · Q1

Gravitational potential & field strength of a uniform planet

(a) [2 marks]

The **work done per unit mass**
in bringing a small test mass from **infinity to the point**.

(b)(i) [2 marks]

The gravitational potential is defined as zero at infinity. Work is done by the two masses as they move towards each other (equivalently, work must be done on them to move them apart), so the potential decreases below zero — hence φ is **negative** for all values of x.

(b)(ii) [2 marks]

$\phi = -GM/r$. At the surface, $r = R$ and $\phi = -\Phi$, so $-\Phi = -GM/R$.
$M = \mathbf{\Phi R/G}$, where G is the (universal) gravitational constant.

(b)(iii) [1 mark]

$g_0 = GM/R^2 = (\Phi R)/R^2 = \mathbf{\Phi/R}$

(b)(iv) [3 marks]

Two curves: one in the **top-left quadrant** (between $x = -3R$ and $x = -R$) and one in the **bottom-right quadrant** (between $x = R$ and $x = 3R$). The upper curve passes through $(-R, g_0)$ and $(-2R, \frac{1}{4}g_0)$, with a non-zero value of g at $x = -3R$; the lower curve passes through $(R, -g_0)$ and $(2R, -\frac{1}{4}g_0)$, with a non-zero value of g at $x = 3R$. Between $-3R$ and $-R$ the curve has a gradient and a magnitude of g that increase towards $-R$; between R and 3R the gradient and the magnitude of g decrease away from R. Nothing is drawn between $x = -R$ and $x = R$.

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$.

P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre, as shown in Figure 1.1.

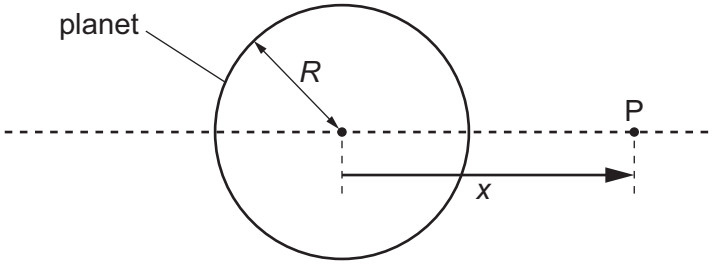


Figure 1.1

Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.

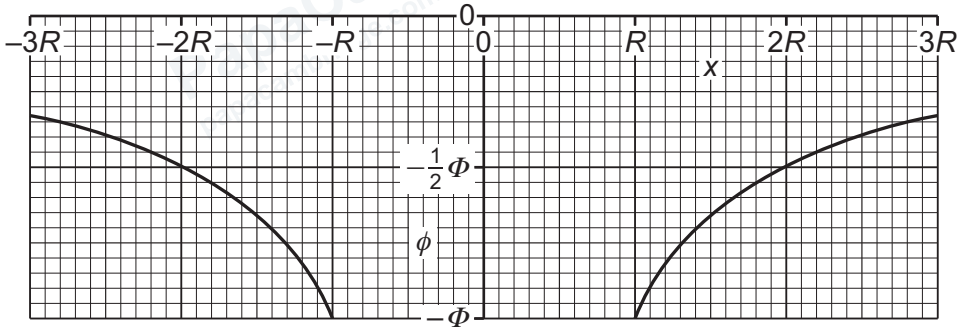


Figure 1.2

(i) Explain why ϕ is negative for all values of x.

.....

.....

..... [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use.

M = ... [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet.

g_0 = ... [1]

(iv) On Figure 1.3, sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do **not** include the region inside the sphere between $x = -R$ and $x = R$.

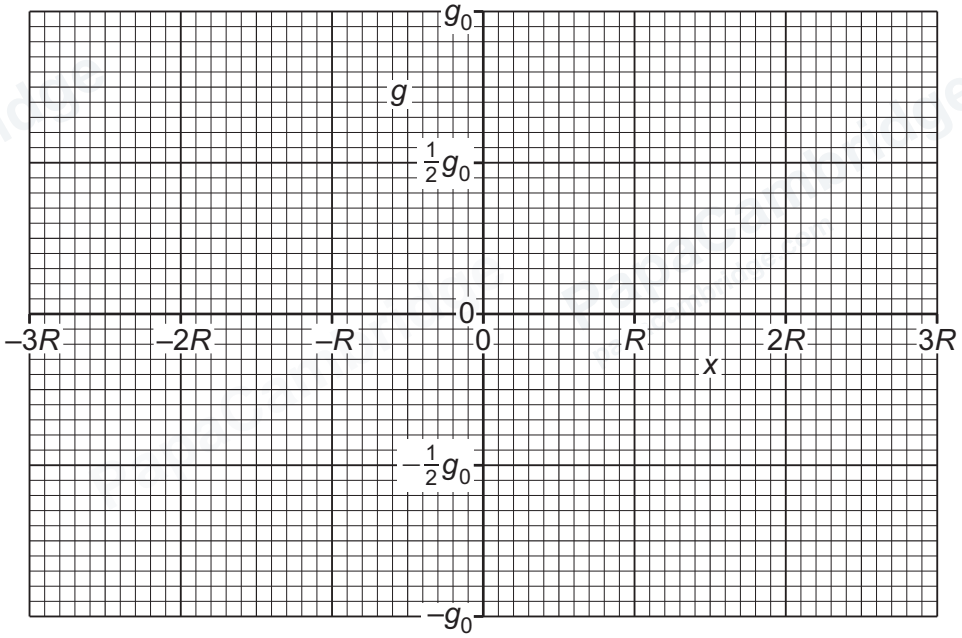


Figure 1.3

[3]

[Total: 10]

Markscheme — 9702/43/M/J/26 · Q1

Gravitational potential & field strength of a uniform planet

(a) [2 marks]

The gravitational potential at a point is the **work done per unit mass** in bringing a small test mass from **infinity** to that point.

(b)(i) [2 marks]

The potential is defined as **zero at infinity**.
Work is done **by** the two masses as they move towards each other from infinity (or work must be done on them to increase their separation), so their gravitational potential energy decreases — hence the potential at any finite point is **negative for all values of x**.

(b)(ii) [2 marks]

At the surface: $\phi = -GM/R = -\Phi$
M = **$\Phi R/G$**, where G is the universal gravitational constant.

(b)(iii) [1 mark]

$g0 = GM/R^2 = G(\Phi R/G)/R^2 =$ **Φ/R**

(b)(iv) [3 marks]

Draw two curves — one in the **top-left quadrant** and one in the **bottom-right quadrant** (nothing drawn between x = −R and x = +R).
Each curve must pass through (±R, ±g0) and (±2R, ±¼g0), with a **non-zero value of g at ±3R**.
The magnitudes of both g and the gradient **decrease** as |x| increases ($g \propto 1/x^2$), so each curve flattens towards the x-axis away from the planet.

2 **(a)** **(i)** Define gravitational field at a point.

.....

..... [1]

(ii) Figure 2.1 shows a horizontal section through a small piece of ground at the Earth's surface and the air at a small distance above it.

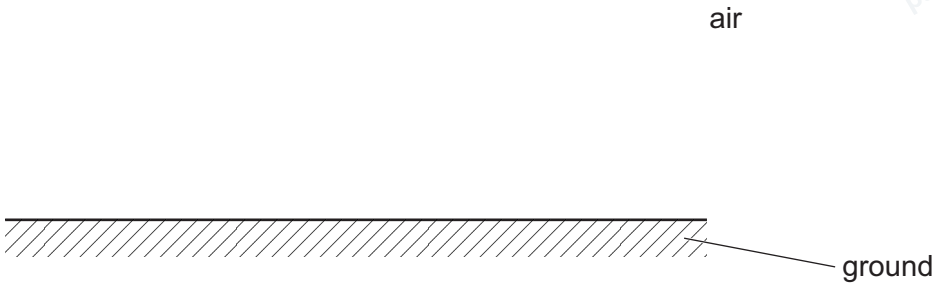


Figure 2.1

On Figure 2.1, draw four field lines to represent the gravitational field of the Earth in the air above the ground. [2]

(b) The Earth may be considered to be an isolated uniform sphere of radius 6.37×10^6 m.

(i) By considering the gravitational field strength at the Earth's surface, calculate a value for the mass of the Earth. Give your answer to three significant figures.

mass = ... kg [2]

(ii) Determine, to three significant figures, the gravitational potential at the Earth's surface. Give a unit with your answer.

gravitational potential = ... unit ..... [2]

(iii) A satellite of mass 96.0 kg is in circular orbit at a height of 1.93×10^6 m above the Earth's surface.

Determine the gravitational potential energy of the satellite.

gravitational potential energy = ... J [2]

[Total: 9]

Markscheme — 9702/42/M/J/26 · Q2

Earth's gravitational field — mass, potential & satellite energy

(a)(i) [1 mark]

The gravitational field (strength) at a point is the **force per unit mass acting on a small test mass placed at that point**.

(a)(ii) [2 marks]

Draw **four straight, parallel field lines perpendicular to the ground** (vertical), approximately evenly spaced, with **arrows pointing downwards towards the ground**.

(b)(i) [2 marks]

$g = GM/R^2 \rightarrow M = gR^2/G = 9.81\times(6.37\times10^6)^2/(6.67\times10^{-11})$
$= \mathbf{5.97\times10^{24}\ kg}$

(b)(ii) [2 marks]

$\varphi = -GM/R = -6.67\times10^{-11}\times5.97\times10^{24}/(6.37\times10^6)$
$= \mathbf{-6.25\times10^7\ J\ kg^{-1}}$

(b)(iii) [2 marks]

Orbital radius $r = (6.37 + 1.93)\times10^6 = 8.30\times10^6$ m
$E_P = -GMm/r = -6.67\times10^{-11}\times5.97\times10^{24}\times96.0/(8.30\times10^6)$
$= \mathbf{-4.61\times10^9\ J}$

2 A satellite is in geostationary orbit around the Earth. The orbit is circular.

(a) (i) State what is meant by a geostationary orbit.

.....

.....

..... [1]

(ii) State **two** properties of the orbit.

1 .....

.....

2 .....

.....

[2]

(b) The orbit has radius R and period T.

(i) Determine an expression, in terms of R and T, for the speed v of the satellite in its orbit.

v = ... [1]

(ii) Use Newton's law of gravitation to show that the mass M of the Earth is given by

$$M = \frac{4\pi^2 R^3}{GT^2}$$

where G is the gravitational constant.

[2]

(c) The Earth may be considered to be a sphere of radius 6370 km.
Geostationary satellites orbit at a height of 35 900 km above the surface of the Earth.

Determine a value, to three significant figures, for the mean density of the Earth.

density = ... $kg\,m^{-3}$ [4]

[Total: 10]

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Markscheme — 9702/44/M/J/26 · Q2

Geostationary satellite — Earth's mass & mean density

(a)(i) [1 mark]

A geostationary orbit is one in which the satellite **remains above the same point on the Earth's surface**.

(a)(ii) [2 marks]

Any two of: the orbit is **equatorial** (directly above the Equator); the satellite orbits **from west to east**; the orbital period is **24 hours** (equal to the Earth's period of rotation).

(b)(i) [1 mark]

$v = \mathbf{2\pi R/T}$

(b)(ii) [2 marks]

$GMm/R^2 = mR\omega^2$ and $\omega = 2\pi/T$:
$GMm/R^2 = mR(2\pi/T)^2 \rightarrow M = \mathbf{4\pi^2R^3/(GT^2)}$

(c) [4 marks]

Orbit radius $R = (6370 + 35\,900)\times10^3 = 4.23\times10^7$ m; $T = 24\times3600 = 8.64\times10^4$ s
$M = 4\pi^2R^3/(GT^2) = 4\pi^2\times(4.23\times10^7)^3/(6.67\times10^{-11}\times(8.64\times10^4)^2) = 5.99\times10^{24}$ kg
$V = (4/3)\pi r^3 = (4/3)\pi\times(6.37\times10^6)^3 = 1.08\times10^{21}$ m^3
density $= M/V = 5.99\times10^{24}/1.08\times10^{21} = \mathbf{5530\ kg\ m^{-3}}$